

MD ASHIQUL ISLAM BHUIYAN

AI / ML Engineer

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SUMMARY

Computer Science and Engineering graduate (B.Sc., 2026) specializing in Artificial Intelligence and Machine Learning, with hands-on experience training GNN models on biomedical data, fine-tuning transformer-based NLP models, and building full-stack systems in Python, Go, and TypeScript. Comfortable taking a model or platform from research through deployment, covering Streamlit apps, FastAPI and Fastify backends, Docker-based infrastructure, and real-time dashboards. Also experienced in post-quantum cryptography implementation and blockchain-based verification systems. Seeking AI/ML and software engineering roles where research translates into deployed, working systems.

EDUCATION

B.Sc. in Computer Science and Engineering

2022 – 2026

Daffodil International University (DIU), Dhaka, Bangladesh

CGPA: 3.73 / 4.00, **Specialization:** Artificial Intelligence and Machine Learning

Thesis: INDIGO, GNN-based Drug-Drug Interaction Prediction

Relevant Coursework: Machine Learning, Deep Learning, Algorithms, Computer Graphics, Cryptography and Network Security, Database Systems

TECHNICAL SKILLS

- Languages:** Python, Go, JavaScript, TypeScript, Dart, PHP
- AI / ML:** PyTorch, PyTorch Geometric, TensorFlow, Scikit-learn, Hugging Face Transformers, Focal Loss, Isolation Forest, LSTM
- Graph ML:** GNN, GCN, GIN, Link Prediction, Node Embeddings, Attention Pooling
- NLP:** BanglaBERT, BERT Fine-tuning, Sequence Classification, Hyperparameter Tuning
- Data and Visualization:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Feature Engineering, Model Validation
- Backend:** FastAPI, Fastify, Node.js, PostgreSQL, MySQL, MongoDB, Redis, BullMQ
- Frontend:** React, Next.js, Vue.js, Tailwind CSS, Flutter, Streamlit, HTML, CSS
- Cryptography:** ML-KEM-1024 (CRYSTALS-Kyber), ML-DSA-87 (CRYSTALS-Dilithium), AES-256-GCM, SHA3-512, liboqs
- Cloud and DevOps:** Docker, Docker Compose, Kubernetes, Oracle Cloud (ARM), Git, GitHub Actions, Google Colab

PROJECTS

INDIGO, Drug-Drug Interaction Prediction via GNN

2025 – Present

Python, PyTorch Geometric, GCN, GIN, DrugBank, Focal Loss, Attention Pooling, RDKit

- Designed a 4-version GNN pipeline (GCN baseline to GIN with triple pooling and multi-head attention) for multi-label DDI prediction across 87 interaction types on DrugBank v5.1, using strict drug-disjoint 70/15/15 splits across 11,107 drugs.
- Achieved ROC-AUC 0.9771 and PR-AUC 0.5410 with the best model (V4 GIN).
- Applied Focal Loss ($\gamma=2$), reducing the false-negative rate by 18% versus a cross-entropy baseline.
- Implemented negative sampling and molecular feature engineering to improve link prediction generalization on unseen drug pairs.
- Deployed as a Streamlit web application.

Bengali Sentiment Classification, BanglaBERT Fine-Tuning

2025 – Present

Python, Hugging Face Transformers, BanglaBERT, PyTorch, NLP

- Fine-tuned csebuetnlp/banglabert on a 3-class Bengali sentiment dataset, achieving 88.37% accuracy and a 0.8846 weighted F1 score.
- Designed a Google Forms data collection pipeline producing 2,500+ labeled Bengali social media posts.
- Conducted a hyperparameter search across learning rates ($1e-5$ to $5e-5$) and epochs (3-10); documented per-class precision, recall, and confusion matrix analysis.

eShakkhor, Post-Quantum Digital Document Signing Platform

2026 – Present

Next.js 14, TypeScript, PostgreSQL, Redis, liboqs WebAssembly, ML-DSA-87, ML-KEM-1024

- Built a PQC document signing platform using ML-DSA-87 (CRYSTALS-Dilithium) and ML-KEM-1024 (CRYSTALS-Kyber) compiled to WebAssembly via liboqs, targeting NIST PQC Level 5 security.
- Integrated steganographic watermarking (AES-256-GCM plus SHA3-512) for tamper-evident embedding, with Redis-backed sessions and BullMQ job queues for signing workflows.

- Reached sub-300ms signing latency for ML-DSA-87 operations in-browser via WASM, with key sizes 80% smaller than RSA-4096 for the same security margin.

MedExPart, Full-Stack Healthcare Platform

2025 – 2026

Next.js, Fastify, TypeScript, Prisma, PostgreSQL, Redis, BullMQ, Docker, Oracle Cloud

- Built a multi-role healthcare platform covering blood donation, emergency SOS, doctor consultations, medicine delivery, and AI health chat across 15+ REST API endpoints.
- Deployed on Oracle Cloud Free Tier ARM via Docker Compose with GitHub Actions CI/CD, and implemented SSE streaming for real-time AI chat via the Anthropic API.
- Designed RBAC authentication with Prisma ORM and JWT session handling, with Redis caching for sub-50ms response times on high-frequency endpoints.

GitQuest, GitHub Gamification Platform

2026

Go, Vue.js, GitHub REST API, PostgreSQL

- Built a GitHub activity gamification platform in Go with a Vue.js frontend, integrating the GitHub REST API to track commits, PRs, and issues and convert them into XP, levels, and achievement badges.
- Implemented concurrent Go routines for parallel GitHub API fetching, reducing data sync time by 60% versus sequential requests.

VeriChert, Blockchain Academic Credential Verification

2025

Solidity, Ethereum Sepolia, IPFS, Merkle Tree, AES-256-GCM, OpenZeppelin

- Deployed Solidity smart contracts on Ethereum Sepolia using Merkle tree batching for gas-efficient credential verification, with IPFS storage secured by AES-256-GCM encryption.
- Reduced per-credential on-chain cost by 74% through Merkle batch proofs versus an individual-transaction approach.

FloraScan, AI Botanical Image Classifier

2025

Python, PyTorch, ResNet34, Transfer Learning, Streamlit, Oxford Flowers102

- Fine-tuned ResNet34 (ImageNet pretrained) on Oxford Flowers102 (8,189 images, 102 classes) with rotation, color jitter, and random scaling augmentation, achieving 91.4% top-1 validation accuracy.
- Deployed as a Streamlit app with Plotly top-3 confidence visualization; CPU inference latency under 120ms per image.
- Public repository: github.com/ASHQEELA/FloraScan

CERTIFICATIONS

- **AWS Academy Graduate, Cloud Foundations**, AWS Academy, April 2026

ADDITIONAL

- **Languages:** Bengali (native), English (professional)
- **Open Source:** github.com/ASHQEELA, public repositories including FloraScan and MedXpert